

LAHORE ALLAMA IQBAL INTL, Pakistan

WMO: 416410

Lat: 31.522N

Lon: 74.404E

Elev: 217

StdP: 98.75

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.2	4.9	-3.2	3.0	18.0	-1.1	3.5	16.6	6.3	15.4	5.5	16.3	0.5	270	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.8	43.2	23.3	41.9	23.3	40.1	23.3	29.1	34.6	28.6	33.8	28.1	33.1	3.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	24.8	32.2	27.2	23.6	31.6	26.9	23.2	31.5	96.9	34.3	94.2	33.9	91.8	33.3	35.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.0	6.4	5.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			1.2	46.0	1.8	1.2	-0.1	46.8	-1.1	47.5	-2.1	48.2	-3.4	49.0		
			0.5	30.3	1.8	1.7	-0.7	31.5	-1.8	32.5	-2.8	33.4	-4.1	34.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.4	12.0	15.9	21.3	27.4	32.3	33.1	31.1	30.5	29.3	25.6	19.3	13.9	(6)
	DBStd	7.70	2.39	2.58	3.28	3.44	3.13	3.17	2.28	1.95	1.96	2.68	2.33	2.49	(7)
	HDD10.0	12	9	1	0	0	0	0	0	0	0	0	0	2	(8)
	HDD18.3	433	197	74	10	0	0	0	0	0	0	0	16	137	(9)
	CDD10.0	5253	71	166	352	523	693	692	653	636	579	484	280	124	(10)
	CDD18.3	2634	0	7	103	274	434	442	395	377	329	226	46	1	(11)
	CDH23.3	36466	3	93	1093	3576	6832	7038	5651	5098	4111	2433	507	31	(12)
	CDH26.7	21313	0	9	425	2095	4612	4760	3271	2749	2114	1176	101	0	(13)
(14) Wind	WSAvg	2.1	1.5	2.1	2.3	2.6	2.9	3.1	2.8	2.3	1.8	1.4	1.2	1.2	(14)
(15) Precipitation	PrecAvg	550	18	19	37	15	19	31	198	156	63	8	2	12	(15)
	PrecMax	900	100	78	154	68	58	85	665	499	164	33	15	82	(16)
	PrecMin	260	0	0	1	0	0	0	43	29	0	0	0	0	(17)
	PrecStd	200	22	20	41	17	18	27	138	127	50	10	4	20	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.2	28.0	35.2	42.0	45.1	45.4	41.1	38.2	37.2	36.1	30.8	25.9	(19)
		MCWB	13.0	16.0	20.0	21.2	23.3	23.3	26.8	27.0	25.3	23.4	17.8	14.6	(20)
	2%	DB	21.8	26.0	33.0	39.9	43.2	43.8	38.9	37.0	36.2	35.0	29.0	24.1	(21)
		MCWB	12.6	15.2	19.4	20.7	22.9	23.6	27.1	27.2	25.4	22.7	17.8	14.2	(22)
	5%	DB	20.1	24.1	31.0	38.0	42.0	42.0	37.1	35.9	35.2	33.9	27.9	22.8	(23)
		MCWB	12.0	14.9	18.7	20.6	22.9	24.0	27.3	27.1	25.5	22.2	17.0	13.6	(24)
	10%	DB	18.8	22.8	29.0	36.0	40.2	40.1	35.9	34.9	34.2	32.6	26.8	21.1	(25)
		MCWB	11.5	14.2	18.0	20.3	22.3	24.5	27.1	27.2	25.4	21.7	16.8	12.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	18.3	22.5	25.1	27.8	29.3	30.2	29.7	28.7	26.1	20.7	16.9	(27)
		MCDB	20.3	24.1	31.2	36.4	40.3	37.5	35.4	34.5	33.6	33.1	26.8	22.2	(28)
	2%	WB	14.1	17.0	21.1	23.1	26.2	28.3	29.2	29.0	27.8	25.0	19.5	15.7	(29)
		MCDB	19.1	23.1	29.6	35.4	38.3	36.3	34.3	33.7	32.8	31.9	25.9	21.4	(30)
	5%	WB	13.1	16.1	20.0	22.2	25.0	27.5	28.7	28.5	27.2	24.0	18.4	14.7	(31)
		MCDB	17.8	21.8	28.1	34.5	37.2	35.5	33.8	32.9	32.3	30.3	25.1	20.4	(32)
	10%	WB	12.3	15.1	19.0	21.2	24.0	26.8	28.2	28.0	26.6	23.2	17.5	13.8	(33)
		MCDB	17.0	20.7	26.9	32.5	37.0	35.1	33.1	32.2	31.6	29.7	24.4	19.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	12.4	13.1	14.4	14.3	11.8	8.3	7.9	9.7	13.2	14.7	14.0	(35)
		MCDBR	15.0	14.5	15.4	16.7	15.8	14.6	10.0	9.2	10.7	13.3	15.5	15.8	(36)
	5% WB	MCWBR	7.3	6.0	5.1	4.9	4.9	4.2	3.5	3.2	3.4	3.9	6.1	7.2	(37)
		MCWBR	12.4	12.1	13.5	14.6	14.2	10.6	8.3	7.9	9.0	11.4	13.6	13.3	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.488	0.495	0.512	0.573	0.654	0.748	0.789	0.697	0.606	0.659	0.622	0.538	(40)
		taud	1.855	1.891	1.880	1.789	1.652	1.527	1.499	1.657	1.788	1.596	1.616	1.755	(41)
	Ebn at Noon	Ebn at Noon	695	747	771	743	687	623	595	644	685	594	563	615	(42)
		Edh at Noon	168	178	193	219	253	286	293	247	209	237	214	178	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.86	3.94	5.25	6.28	6.94	6.45	5.48	5.28	5.20	4.60	3.54	2.95	(44)
	RadStd	RadStd	0.33	0.41	0.33	0.28	0.24	0.37	0.34	0.40	0.28	0.23	0.19	0.18	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page